

Piter Z. Garcia Bautista

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Research Summary

Graduate researcher and NSF Robert Noyce Scholar working at the intersection of quantum computing, compiler-level optimization, and reproducible experimentation. Current research spans quantum network optimization, fault-tolerant routing, and fairness-aware decision workflows. Specifically interested in compiler-assisted precision and resource management for fault-tolerant quantum workloads, as explored in the SPQR project and TAFFO-inspired frameworks for quantum error correction code selection.

Education

Rochester Institute of Technology

Expected Aug 2026

Master of Science, Data Science — GPA 3.9

Rochester, NY

- Concentration: Bioinformatics & Artificial Intelligence.
- Relevant coursework: Foundations of Data Science (DSCI-633), Software Engineering for Data Science (DSCI-644), Applied Data Science I (DSCI-601), Bioinformatics Algorithms (BIOL-630), Data-Driven Knowledge Discovery (ISTE-780).

University of Rochester, Warner School of Education

Expected Aug 2026

Master of Science track, Teaching Computer Science K-12 — GPA 4.0

Rochester, NY

- NSF Robert Noyce Teacher Scholarship Program (\$30,000 full funding).
- Advanced Certificate in Leadership in Disability and Inclusive Practices.

Rochester Institute of Technology

2015

Master of Science, Computer Science

Rochester, NY

- Concentrations: Computer Graphics, Artificial Intelligence, Big Data Analytics.
- Coursework in compiler design, algorithm analysis, and distributed systems.

Farmingdale State College

2012

Bachelor of Science, Computer Engineering Technology — GPA 3.3

Farmingdale, NY

- Dual degree: Electrical Engineering & Computer Engineering Technology.
- Honors: CSTEP Award & Merit, Dual BS Scholarship.

Research Experience & Projects

Rochester Institute of Technology

Fall 2025 – Present

Graduate Assistant, Quantum and AI Research

Rochester, NY

- Conduct research on quantum network path optimization and fault-tolerant routing under adversarial and stochastic conditions.
- Develop compiler-level experimental infrastructure for quantum circuit workloads using Qiskit and Cirq.
- Implement and validate quantum error correction code selection workflows in simulation (Stim-compatible pipelines).

- Apply LLVM/MLIR IR-level analysis concepts to quantum compilation pipeline studies; reviewed TAFFO precision-tuning methodology for applicability to quantum resource management.

RIT & University of Rochester

2024 – Present

Lead Researcher, EQUITAS Framework for Diagnostic Justice

Rochester, NY

- Developed an interdisciplinary framework integrating fairness-aware ML, bioinformatics, and trauma-informed analysis.
- Built reproducible bias-assessment workflows and reported equity gaps across algorithmic variants.
- Coordinated cross-domain collaboration across computing, education, and health research contexts.

Rochester Institute of Technology

2024 – Present

Principal Investigator, Equitable Bioinformatics Project (ISTE-780)

Rochester, NY

- Applied systematic fairness analysis to RNA secondary structure prediction pipelines.
- Implemented reproducible evaluation codebase with automated metric reporting and statistical testing.

Teaching & Mentoring Experience

University of Rochester, Warner School of Education

Sep 2024 – Present

CS Teacher Candidate, K–12 Computer Science Teaching Placement

Rochester, NY

- Deliver K–12 computer science instruction in public school placement as part of NSF Noyce Scholar program.
- Design and teach CS curriculum aligned with NYS learning standards; mentor students in computational thinking and algorithmic reasoning.

Varsity Tutors & Independent Practice

2021 – Present

STEM Tutor and Academic Mentor

Remote & In-Person

- Tutor Computer Science, Data Science, Mathematics, and Statistics for diverse learners.
- Design culturally responsive instruction and mentoring support for underrepresented students in STEM.

Rochester Institute of Technology

2013 – 2015

Faculty Assistant, Graduate Office

Rochester, NY

- Supported graduate program operations, student documentation, and faculty coordination.

Selected Publications & Academic Contributions

- *Quantum Network Path Optimization and Fault-Tolerant Routing* (manuscript under review, 2025).
- *Designing Equitable AI Systems for Diagnostics: An IDAI700 Research Portfolio* (Project Report, 2025).
- *Equitable Bioinformatics: Enhancing Diagnostic Decision-Making through RNA and Biomarker Data* (ISTE-780 Project Report, 2025).
- *Trauma-Informed Bioinformatics for Diagnostic Justice: An EQUITAS Framework* (ED-447 Capstone, 2025).
- *RNA Secondary Structure Prediction and Visualization: Implementing the Nussinov Algorithm with MFE Enhancements* (BIO-630 Final Project, 2025).

- *Fairness-Aware Bandits for Clinical Decision Systems* (DSCI-601 Project Report, 2023).
- *ASL Recognition Application: Image Processing for Sign Language Translation* (MS Thesis, 2015).

Technical Skills

Programming: Python, R, Java, C++, C#, MATLAB, SQL, JavaScript, Bash

Quantum Computing: Qiskit, Cirq (gate-based quantum circuit simulation and algorithm development)

Compiler & Systems: LLVM/MLIR compiler backend framework; TAFFO/Precimonious (approximate computing and precision tuning)

Machine Learning: PyTorch, TensorFlow, scikit-learn, XGBoost, fairness-aware ML, contextual bandit workflows

Data & Analysis: pandas, NumPy, SciPy, SHAP analysis, statistical testing, reproducible benchmarking

Infrastructure: AWS, GCP, Docker, Git/GitHub, Linux, LaTeX

Professional Experience

VEDADATA Inc.

Data Solutions Engineer

Sep 2022 – Aug 2024

New York, NY

- Designed and implemented data pipelines and analytics workflows using Python and cloud services.

VIOME Inc.

AI Data Solutions Engineer

Jan 2021 – Sep 2022

New York, NY

- Built ML workflows for healthcare applications with production deployment and monitoring support.

Selected Fit for NTNU Quantum Compiler Technologies Position

- Directly relevant research in quantum circuit optimization and fault-tolerant routing; active study of compiler-assisted precision tuning methodologies (TAFFO, Precimonious) and their extension to QEC code selection.
- Hands-on experience with LLVM/MLIR IR-level analysis and quantum circuit frameworks (Qiskit, Cirq) providing a practical foundation for compiler backend contributions within the SPQR project.
- Reproducibility-first experimental discipline (multi-environment: local, Colab, GCP) with traceable infrastructure—well matched to SSG’s open-source and benchmark-driven research culture (CGO, PLDI).
- Interest in PoliMi collaboration; familiar with TAFFO’s Value Range Analysis approach as the classical precedent the SPQR project extends to the quantum domain.